

WASP-12b

R-1515

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_190110 · created 2026-08-01T16:28:55+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458493.74918 +/- 0.00058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.044
Transit depth (Rp/Rs)^2	2.03 +/- 1.25 [%]
Orbital Inclination (inc)	86.69 +/- 2.85
Airmass coefficient 1 (a1)	0.95 +/- 0.048
Airmass coefficient 2 (a2)	0.05 +/- 0.15
Scatter in the residuals of the lightcurve fit is	5.13 %
Best Comparison Star	#1 - [448, 138]
Optimal Aperture	1.88
Optimal Annulus	7.16
Transit Duration (day)	0.1309 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.044 vs 0.1178 (deviation 0.55σ)
Duration fit vs published	0.1309 d vs 0.125 d (deviation 0.95σ)
Mid-time O–C	-1.4 min
Depth SNR	3.2
Residual scatter	5.13 %

Light curve

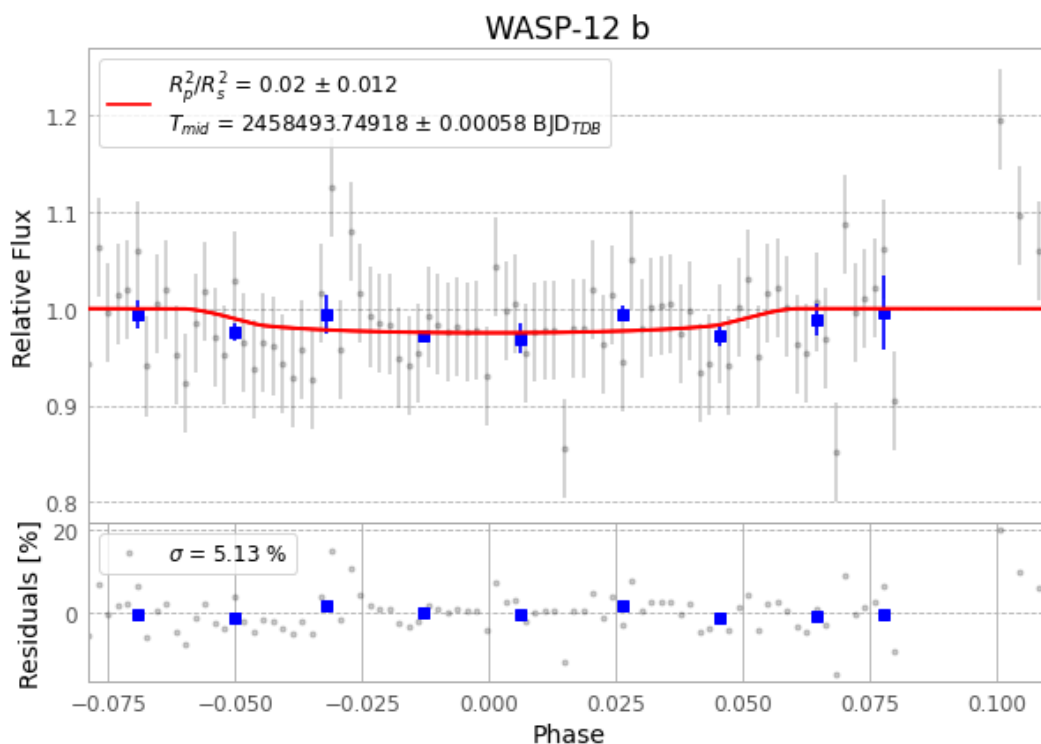


Figure 1 — FinalLightCurve_WASP-12 b_2019-01-09.png (SHA-256 2a4295628ea82555...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c4ad47a3f5d5d5fc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

100 FITS frames (66 MB), WASP-12190110034512.FITS → WASP-12190110084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190110.log (ee55520c057c...), FinalLightCurve_WASP-12 b_2019-01-09.png (2a4295628ea8...), FinalLightCurve_WASP-12 b_2019-01-09.csv (1696a615d7a2...)

Compute resources

Wall time 511.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 16:28Z.